

FOREIGN PAYLOADS

CIXS



From the UK, meant for mapping abundance of minerals such as magnesium and aluminum on the moon

SARA



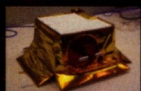
A Swedish venture to study the way moon's surface reacts to solar wind

Mini SAR



NASA of US is using this instrument to detect water ice in the permanent shadow areas at the lunar poles

SIR-2



From Max Planck Institute of Germany, to map the mineral resources and the layering of moon's crust

RADOM



Bulgaria's payload to help study radiation dose map of space near the moon at various altitudes

M3



Another NASA payload to map lunar mineral resources to help support future targeted missions

INDIAN PAYLOADS

HySI



This CCD camera will help map the minerals on the moon's surface with high resolution images

HEX



Using very high energy X-rays, this will also help scientists identify water ice deposits in the polar regions of the moon

LLRI



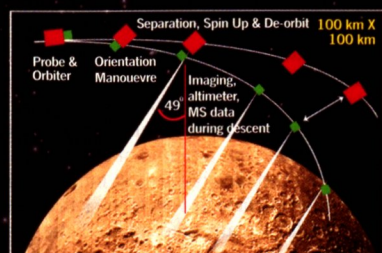
It will help determine the accurate altitude of Chandrayaan and the internal structure of the moon

TMC



Using a high resolution camera of 5m it would map the entire topography of the moon and its gravitation field

MIP The objective of the Moon Impact Probe is to help understand the technology for landing a vehicle on a desired location on the moon



vehicle's performance. With Doordarshan telecasting the lift-off live, the country too had tuned into this prestigious launch. After all, ISRO was embarking on a mission that no Indian had ever done before—a journey to the moon 4,00,000 km away, almost 34 times the distance from Delhi to New York. Eighteen agonising minutes later, the launcher injected Chandrayaan-1 into an elliptical orbit of 36,000 km around the earth at its apogee (farthest point from the earth) and 250 km at its perigee (closest point).

Bristling with scientific instruments put together by seven nations including India, Chandrayaan-1 was finally on its lonely way. Despite travelling at 30 times the speed of sound because of its elliptical orbit, it would still take 22 days before it could have a rendezvous with the moon and break free from the earth's gravity. Soon after the successful lift-off, ISRO Chairman G. Madhavan Nair told INDIA TODAY: "It's a giant leap for India and ISRO. If the launch was complex, the days ahead would prove most challenging for us." (see interview)

That's because ISRO is entering uncharted seas. For the first time it will operate out of earth's gravitational field. Now they would have to take care of perturbations coming from planetary movements, solar flares and the gravity flux caused by a constantly rotating moon. Commanded through signals from the largest space antenna ever built by ISRO in Byalalu near Bangalore, Chandrayaan would begin a complex set of manoeuvres that would gradually increase its orbit in the days ahead. In a series of five controlled firings of its on-board engine spread over a week, it would raise its apogee to 3,80,000 km bringing it 500 km from the moon. Then it would retrofire its rockets to lower its altitude and be captured by moon's gravity before obtaining a stable circular orbit of 100 km around the moon.

Nair warns that if the angle of descent deviates by a fraction or the velocity is incorrect, it would send Chandrayaan crashing into the moon or into deep space from where it can't be recovered. When the US sent orbiters to the moon in the '60s, the first six launches failed. The Soviets who sent 29 missions from 1959 to 1976 before stopping altogether reported nine failures. Conscious of the difficult task ahead, M. Annadurai, Chandrayaan's soft-spoken